

Arkodeep Chatterjee

Full-Stack AI Engineer · Realtime Voice Agents, Agentic Systems & Production LLM Infrastructure
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EXPERIENCE

• Neosophical Labs

Backend AI Engineer

Remote

Feb 2026 – Present

- Built a LangGraph multi-agent ingestion pipeline auto-classifying and mapping Excel, XML, and PDF payroll data into a unified Postgres schema; cut per-client HR migration from 3 weeks to under 2 hours
- Engineered tool-calling agents for header-to-column mapping (OpenRouter LLMs) and Gemini VLM-driven PDF→JSON extraction; ingested 27K+ employee records across 14+ enterprise clients at 97% mapping accuracy
- Shipped AI personalization backend for Ahoum Seeker (wellness companion), powering practitioner-seeker matching for 3K+ users with <300ms p95 latency

• Emvo AI

Founding Realtime AI Engineer

Remote

May 2025 – Nov 2025

- Architected “Agentverse,” a sub-second realtime voice engine delivering <800ms p95 response across 50+ enterprise integrations
- Built modular inference pipelines that collapsed client onboarding from weeks to under 30 minutes, powering 8K+ daily voice sessions
- Cut inference cost 34% via streaming + concurrency optimizations while sustaining 98%+ uptime under multi-tenant production load

• Numly Inc

AI Engineer Consultant

Remote

Nov 2024 – May 2025

- Led Engage AI, an enterprise coaching agent shipped to 6 paying clients; drove a 41% lift in employee active-engagement metrics
- Designed adaptive learning loop personalizing leadership tracks for 2K+ users, cutting average skill-gap closure time by 27%
- Shipped analytics layer surfacing skill drift and coaching ROI, reducing manager onboarding time by 40%

• NexaWings International

CTO

On-Site

Nov 2023 – May 2024

- Led 7-engineer team building AI SaaS products (chatbots, workflow automation, AI integrations) shipped to 12+ industry clients
- Drove \$12K+ in revenue through custom GenAI deployments; raised average client process efficiency 3–4x

PROJECTS

• Dodge – SAP Order-to-Cash Knowledge Graph + Conversational AI

- Built full-stack agentic system unifying 21K+ SAP entities into an interactive force-directed knowledge graph
- Engineered ReAct agent with 6 specialized tools, 3-tier guardrails, conversation memory, and pronoun resolution across multi-turn flows
- Implemented SSE streaming + automatic model fallback on rate limits; FastAPI, LangChain, MongoDB Atlas, Next.js 16

• Lawyer GPT – AI legal assistant for law firms and students

- RAG pipeline trained on contract corpus, generating personalized legal drafts in a single prompt
- Cut legal research time 60% in beta by retrieving semantically similar past cases on demand

• TruePal – Personalized AI companion app

- Adaptive AI delivering empathetic, evolving conversations; scaled beta to 500+ users
- Built customizable avatar system and low-latency secure chat infra for stress-reduction use cases

• Orom – All-in-one SaaS suite for SMBs and entrepreneurs

- Consolidated 6+ marketing tools (waitlists, testimonials, newsletters, popups, email automations) into a single dashboard
- Reduced SMB tool sprawl and subscription cost via unified, embeddable widgets across client websites

SKILLS

- **Programming:** Python, TypeScript, JavaScript
- **AI & LLMs:** LangChain, LangGraph, OpenAI, Claude Agent SDK, Gemini Live API, Google ADK, AWS Bedrock, Azure AI, RAG, vector DBs, PandasAI, n8n
- **Backend & APIs:** FastAPI, Node.js, Express.js, Flask, REST, WebSockets, WebRTC
- **Frontend & Mobile:** React, Next.js, Tailwind, React Native, Expo, Nativewind
- **Infra & DevOps:** AWS, Docker, ECS, Redis, AWS OpenSearch, GitHub Actions, CI/CD
- **Databases:** Postgres, MongoDB, Prisma, Mongoose

ACHIEVEMENTS

- Startup Mahakumbh (Bharat Mandapam, New Delhi) – Only startup from West Bengal selected to participate
- RPV Innovation Challenge, IIT Madras – 2nd prize, Autonomous UAV category
- Global Entrepreneurship Summit, IIT Kharagpur – 2nd prize, Business Plan discipline

EDUCATION

B.Tech, Computer Science and Engineering (AI) – Institute of Engineering and Management (IEM), Kolkata · GPA: 8.78